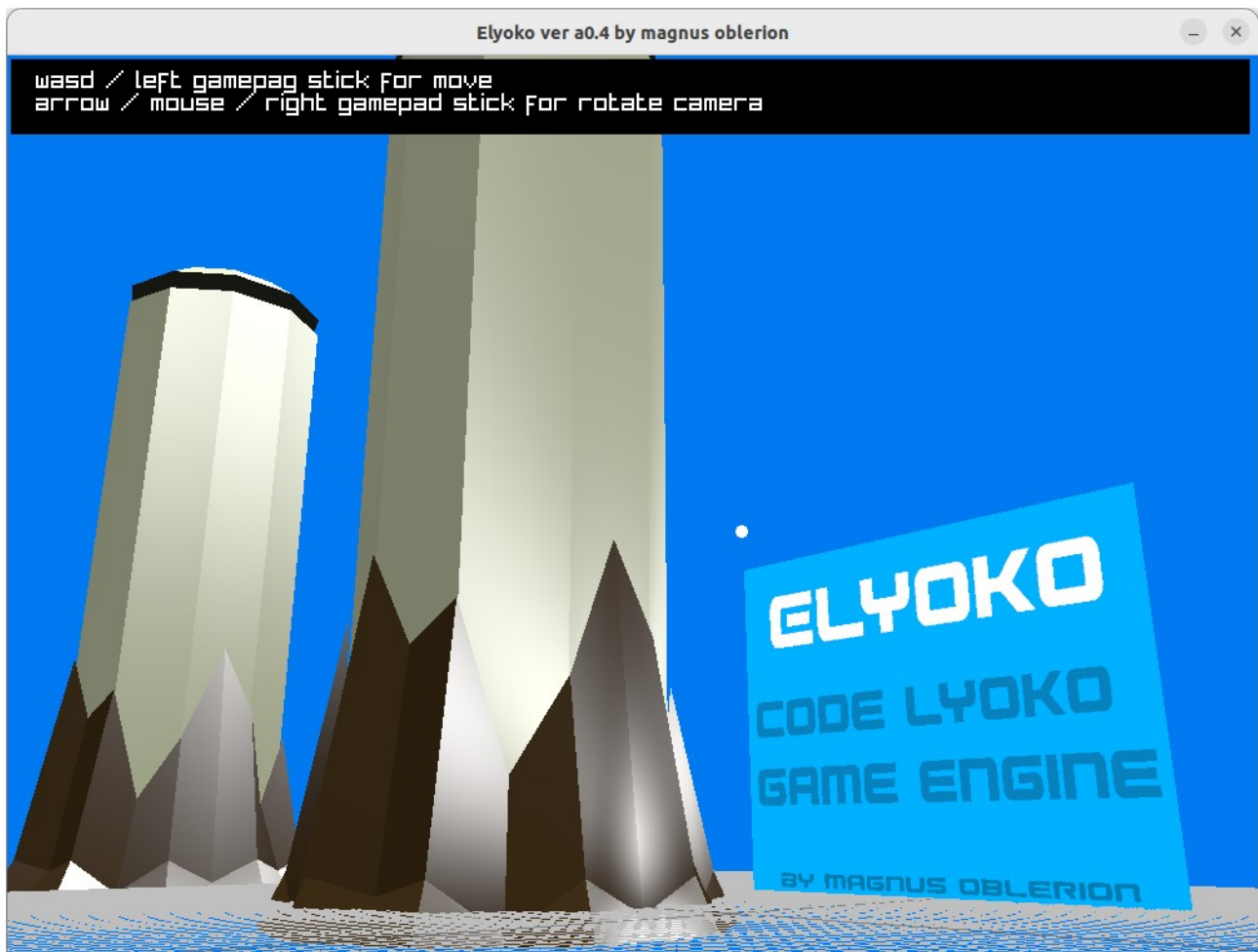


Elyoko

The 3d game engine for
make your lyoko world



by Magnus Oblerion

Description

The 3d game engine make with C , raylib.h and lua.h.

It is lightweight, powerful and easy to extends (lua script).

Features

- Max 100 preload model
- Max 100 preload texture
- load/draw/unload model (obj,gltf,glb)
- load/draw/unload texture (png,jpg)
- load/draw textured plane from image
- draw 3d primitive (cube,sphere)
- draw 2d primitive (rect,circle,point)
- draw text
- move/rotate camera
- keyboard/mouse input -> camera control
- gamepad input -> camera control

Run script CLI

on window

```
elyoko.exe main.lua
```

on linux

```
./elyoko main.lua
```

Run projet folder CLI

Projet folder need main.lua.

on window

elyoko.exe folder

on linux

./elyoko folder

Entry point

If you create file project.entry beside of binary,
elyoko will run project.lua at start.
Or run project folder if exist.

Your *.lua/folder and *.entry need same name for work.

API

1) Main loop

```
function ELYOK02D()  
-- loop 2d  
end
```

```
function ELYOK03D()  
-- loop 3d  
end
```

2) delta time

```
local dt = deltatime() -- get time between 2 frame
```

3) color

```
local id_color = color(255,0,255,20) -- color r=255,g=0,b=255,a=20  
local id_color = color(255,0,200) -- color r=255,g=0,b=200,a=255
```

4) camera 3d

get camera X

```
local camx = camerax()
```

get camera Y

`local` camy = cameray()

get camera Z

`local` camz = cameraz()

camera move

`cameramove(x, y, z)` -- add value to camera position

camera set position

`camerasetpos(x, y, z)` -- set value to camera position

camera rotate

`camerarotate(rotx, roty, rotz)` -- rotation in radiant

Camera set target

`camerasettarget(x, y, z)` -- set points than camera look

camera lock

`cameralock(state)`

-- state=true player can't move/rotate camera

-- state=false player can move/rotate camera

5) Asset loading

Outside loop / protect for run once

load model

`loadmodel(path)` -- path is string

load texture

`loadtexture(path)` -- path is string

load plane textured

`loadplanetexture(texture_name.ext)` -- create model plane with loaded texture

6) Input

btn / btnp

```
if btn(0) then
    -- key w is down or gamepad left stick up
elseif btnp(1) then
    -- key s is pressed or gamepad left stick down
end
```

btn(id) : key is down

btnp(id) : key is pressed

id=0 → key w / gamepad left stick up

id=1 → key s / gamepad left stick down

id=2 → key a / gamepad left stick left

id=3 → key d / gamepad left stick right

id=4 → key x / gamepad button x

id=5 → key c / gamepad button a

mouse

```
local x,y,btnl,btnm,btnr = mouse()
-- x = x mouse
-- y = y mouse
-- btnl = true/false mouse left button
-- btnm = true/false mouse mid button
-- btnr = true/false mouse right button
```

7) 2d

work only in ELYOKO2D

draw pixel

`pix(x, y, color())`

draw fill rectangle

`rect(x, y, width, height, color())`

draw line rectangle

`rectb(x, y, width, height, color())`

draw fill circle

`circle(x, y, radius, color())`

draw line circle

`circleb(x, y, radius, color())`

draw text

`text(str, x, y, scale, color())`

unload texture

`delttexture(name.ext)`

draw texture

`drawtexture(name.ext, x, y)`

8) 3d

work only in ELYOKO3D()

draw cube

cube(x, y, z, width, height, depth, color())

draw sphere

sphere(x, y, z, radius, color())

unload model

delmodel(name.ext) -- name.ext is name without path

draw model

drawmodel(name.ext, x, y, z, rot_x, rot_y, rot_z, scale)
-- rotation in radian
-- scale > 0 , float multiply scale of model

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